

# ernie

A daily visit that notices a bad day at home before it becomes a hospital day.

Christopher Pluntke · September 2026



# Margaret is 84.

Every morning she signs Ernie's book. This morning she was slower, and nobody was there to notice.

Hypoactive delirium, the most common kind, looks like nothing at all: quieter, slower, a little withdrawn. The cause is usually cheap to fix on the day: an infection, dehydration, a new tablet. Missed, it ends in the emergency room.

**60–75%**

of cases unrecognised in hospital and in the ED <sup>1</sup>

**7 M+**

hospitalised US adults affected each year <sup>2</sup>

*Delirium is a change from your own baseline. The home never had one.*

<sup>1</sup> Unrecognised in up to 60% of hospital cases and ~75% in the ED: Delirium in Hospitalized Older Adults, narrative review (PMC 2026); Han et al., ED delirium screening (PMC 2024). <sup>2</sup> Frontiers in Public Health 2026, Medicare Current Beneficiary Survey 2019–21.

[Start again](#)

Step 1 of 4: Signing in

**Thank you,  
Margaret.**

✓ **Done**

Lovely to see you. Here is your page in the book.



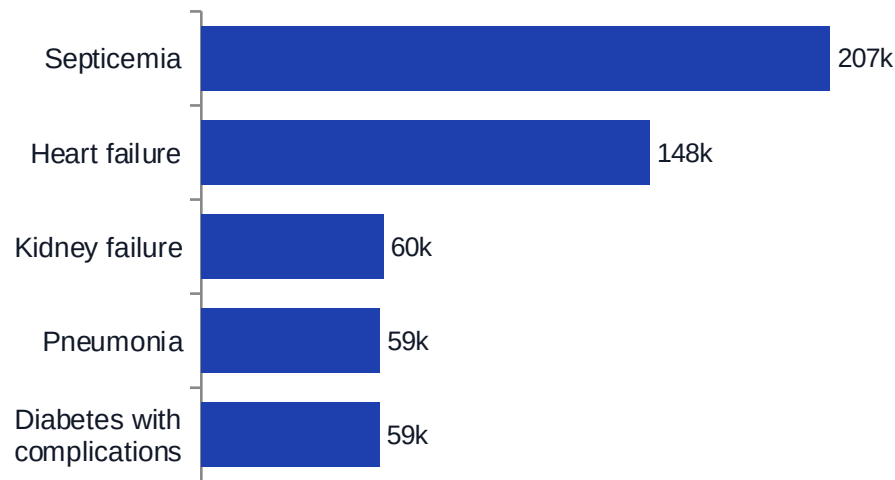
**Thursday 17 September**

**Carry on**

# The gap is the 30 days at home.

Value-based care makes hospitals carry the cost of patients who come back. What sends Medicare patients back, and what a missed bad day costs.

Medicare 30-day readmissions by cause at first admission, 2020 <sup>3</sup>



Delirium is not a diagnosis category here. It rides inside these stays and multiplies the odds of coming back by 2.6 <sup>5</sup>. About 11% of Medicare hospital stays involve it <sup>2</sup>.

**\$36 B**

a year: 2.0 M Medicare readmissions at \$18,100 each, 17 per 100 stays <sup>3</sup>

**up to 3%**

off every Medicare inpatient payment for a year when a hospital's readmissions run high. FY2022: 2,499 hospitals, \$521 M <sup>4</sup>

**+\$8,110**

in hospital charges when a stay involves delirium, against a stay that does not <sup>2</sup>

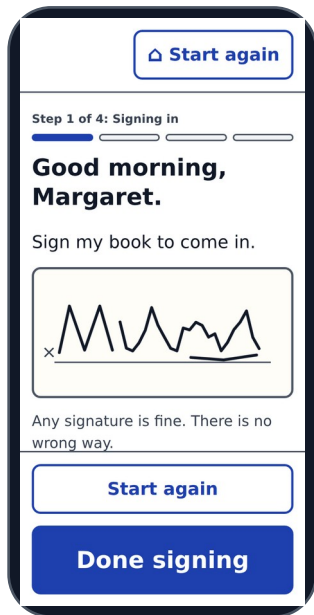
**~27%**

of readmissions judged avoidable (median of 34 studies). Delirium itself: 30–40% preventable <sup>6</sup>

<sup>3</sup> AHRQ HCUP Statistical Brief 307, Nationwide Readmissions Database 2020. <sup>4</sup> CMS Hospital Readmissions Reduction Program; KFF, 10 Years of Hospital Readmissions Penalties (FY2022). <sup>5</sup> Community hospital cohort, n = 8,645, OR 2.60 (PMC 2019). <sup>6</sup> van Walraven et al., CMAJ 2011; Inouye et al., NEJM 1999 (HELP).

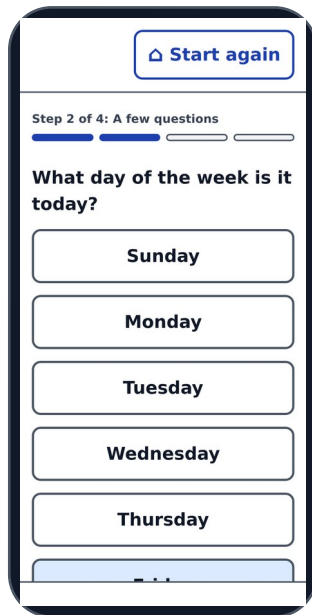
# No tests. Just a visit.

She makes something every day: a signature, her own words, a finished picture. Each one is also a measurement.



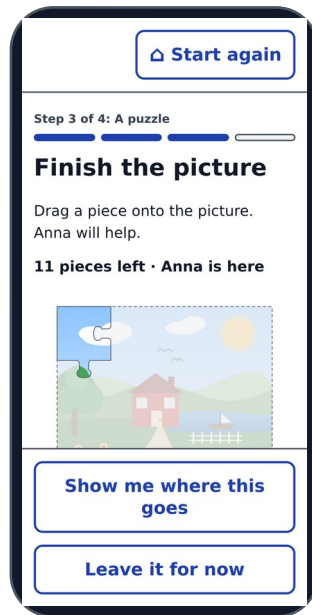
**Sign the book**

motor baseline · orientation



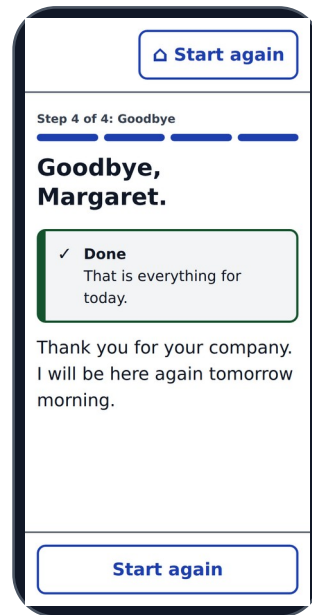
**A few questions**

her own words · latency



**A puzzle, with Anna**

attention · space · company



**Goodbye**

did she come · how fast

# What the phone saw.

*“Margaret’s signature was slower, more hesitant and a different shape than usual today.”*



**Her usual, and today. It looks the same. It wasn't.**

What the family sees. Margaret sees a thank-you and her page in the book.

| FEATURE              | TODAY         | USUAL         | ROBUST Z             |
|----------------------|---------------|---------------|----------------------|
| Time to first stroke | 418 ms        | 420 ms        | -0.1                 |
| <b>Total time</b>    | <b>4.46 s</b> | <b>3.06 s</b> | <b>+5.7 unusual</b>  |
| <b>In-air time</b>   | <b>1.07 s</b> | <b>267 ms</b> | <b>+37.8 unusual</b> |
| Strokes              | 3             | 3             | +0.0                 |
| <b>Mean speed</b>    | <b>0.650</b>  | <b>0.778</b>  | <b>-2.1 unusual</b>  |

Nine features against her own five-day baseline: time to start, total time, pauses, strokes, speed, smoothness, size, shape.

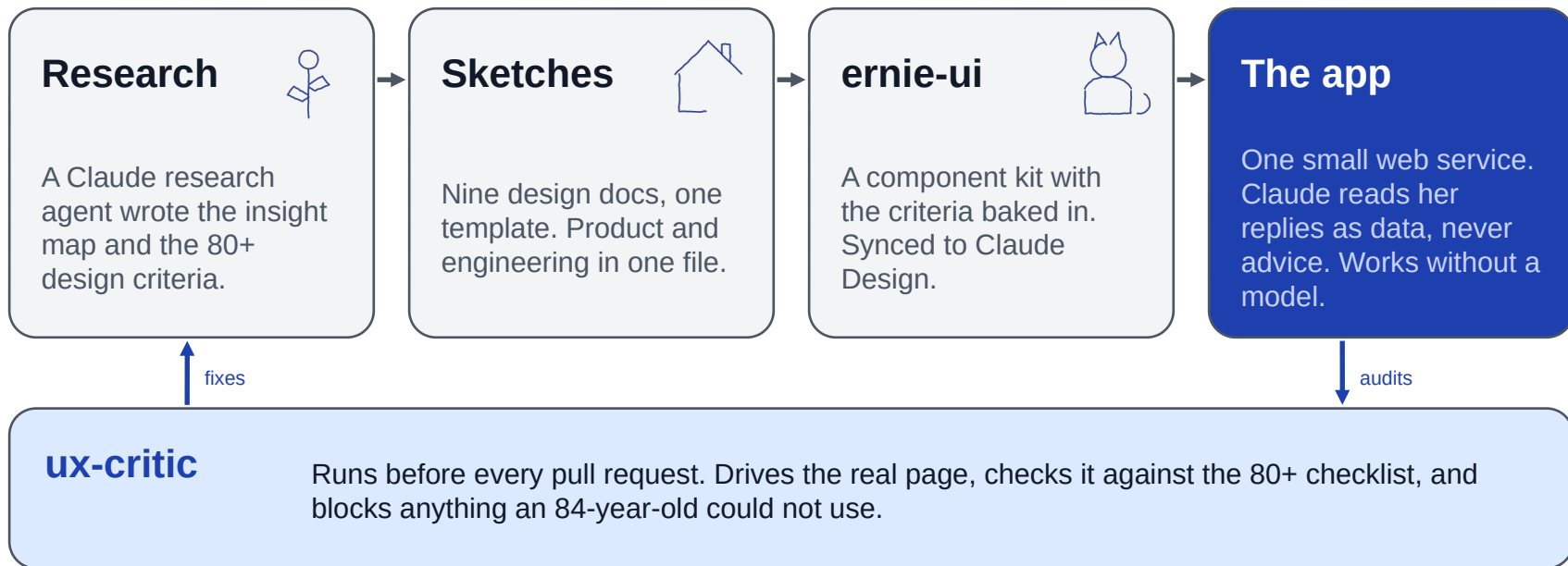
Robust z: today minus her usual, over her usual spread. Well outside on two features is a flag. Three days is a trend.

*Real numbers from the demo build.*



# How it was built.

A pipeline with a reviewer in it.



All of it is in the repository.

# What's next

1

## Data leaves the phone

One process, one file: events, baselines, a family link.

2

## Three real people, three days

Does a finger signature hold a baseline? Does a live doodle beat a check-in?

3

## The explanation layer

A flag opens what changed, what it usually means, who to call before the ER.

*Decision support, never a diagnosis. A human on every flag.*



*Synthetic people only. Not a medical device.*

# Sources

- 1** Delirium unrecognised in up to 60% of hospital cases and about 75% in the emergency department.

*Delirium in Hospitalized Older Adults: A Narrative Review (PMC, 2026); Delirium screening in the emergency department (PMC, 2024).*
- 2** Delirium affects over 7 million hospitalised US adults a year; 11% of Medicare beneficiaries had an episode; +\$8,110 hospital charges per delirium stay (95% CI \$1,860–\$14,360).

*Additional hospitalization costs associated with delirium among older adults, Frontiers in Public Health, 2026 (Medicare Current Beneficiary Survey 2019–2021).*
- 3** Medicare: 2,002,700 30-day readmissions in 2020, 17.0 per 100 index stays, \$36.2 B aggregate, \$18,100 average. Top causes at index admission: septicemia 207,300; heart failure 147,800; acute renal failure 60,000; pneumonia 58,900; diabetes with complications 58,700.

*AHRQ HCUP Statistical Brief #307, Nationwide Readmissions Database, 2020.*
- 4** HRRP reduces a hospital's Medicare inpatient payments by up to 3% for a fiscal year when 30-day readmissions for six conditions exceed the expected rate. FY2022: 2,499 of 3,139 hospitals penalised, average 0.64%, 39 at the 3% maximum, about \$521 M withheld.

*CMS, Hospital Readmissions Reduction Program; KFF, 10 Years of Hospital Readmissions Penalties.*
- 5** Inpatient delirium associated with 30-day readmission, adjusted OR 2.60 (95% CI 1.96–3.44); 718 delirious vs 7,927 non-delirious patients, one community hospital, 2010–2015.

*Association between Inpatient Delirium and Hospital Readmission in Patients ≥65 (PMC, 2019).*
- 6** Median 27% of readmissions judged avoidable across 34 studies (range 5–79%). Multicomponent prevention (HELP) cut delirium incidence from 15.0% to 9.9%, about a third to 40%.

*van Walraven et al., CMAJ 2011; Inouye et al., NEJM 1999.*